

Exploring Auditory and Visual Feet Guidance in Indoor Climbing: Proprioception, Spatial Awareness, and Experience

Abstract

Recent work in SportsHCI has called for a shift from performance-centered evaluation toward a closer examination of embodied experience in sport. This study investigates how different feet-guidance modalities—visual and auditory—influence proprioception, spatial awareness, and climbing experience during indoor bouldering. Using a within-subjects mixed-methods design, twelve amateur climbers completed two bouldering routes under visual laser-based guidance and spatial audio-based guidance.

The results indicate that auditory guidance does not enhance proprioception or spatial awareness in a directly measurable way compared to visual guidance. However, auditory cues encouraged exploratory movement, heightened awareness of foot movement, and a stronger sense of autonomy and flow-related engagement, while also introducing higher cognitive and physical demands. Visual guidance, by contrast, supported more efficient movement planning and lower workload but rendered bodily sensations more implicit.

This study contributes to SportsHCI research on experience-oriented interaction design and highlights the potential of auditory guidance to support exploratory and experience-oriented interaction design in climbing.

Learning reflection

This project largely met the learning objectives outlined in the initial proposal. Through literature review, system design, and a mixed-methods user study, I gained a deeper understanding of how multimodal interaction technologies operate in sports contexts. An important personal learning outcome was developing the ability to abstract a concrete, real-world movement problem into a controlled academic experiment, including defining research questions, simplifying the task without losing its experiential core, and selecting appropriate evaluation methods.

Beyond the original objectives, the project also deepened my understanding of the research process itself. The results did not fully align with my initial expectations, particularly regarding measurable improvements in proprioception and spatial awareness. However, this became a valuable learning experience, reinforcing that unexpected or non-confirmatory findings are a normal and meaningful part of academic research. Rather than treating this as a failure, I learned the importance of examining why such results emerged and providing reasoned interpretations grounded in both practice and theory. If I were to conduct the project again, I would follow a similar overall structure while allocating more time for system

iteration and participant training to better capture embodied and exploratory aspects of climbing.

Introduction

Research in Human–Computer Interaction has increasingly explored how multimodal interfaces can support learning, training, and performance monitoring in physically demanding activities (Elvitigala et al., 2024; Mencarini et al., 2019; Sigrist et al., 2012). Within SportsHCI, recent discussions have explicitly called for a shift away from purely performance-oriented metrics toward experience-centered perspectives that foreground embodied engagement, exploration, and diverse personal goals (Elvitigala et al., 2024). This shift is arguably relevant for recreational and amateur sports contexts, where enjoyment, curiosity, and self-directed exploration often matter as much as measurable performance outcomes.

Climbing provides an arguably suitable context for examining these questions. In recent years, indoor climbing, especially bouldering, has grown rapidly as a recreational activity, attracting a large population of amateur participants for whom enjoyment, exploration, and personal challenge are central motivations rather than competitive performance alone (Climbing Gym Market Size & Share Report, 2025 – 2034, n.d.). Successful climbing (or bouldering) depends not only on physical strength or visual planning, but also on continuous bodily sensing and spatial judgment (Orth et al., 2016). Guidance systems that intervene at this level may therefore shape how climbers attend to their bodies, perceive space, and engage with the activity as a whole. Findings from a formative study suggested that climbers regard foot positioning to be critical for navigation and stability, motivating this study to focus on the investigation of foot-based guidance.

Building on this motivation, the present study investigates how different feet-guidance modalities, namely visual and auditory, shape climbers' proprioception, spatial awareness, and climbing experience. Rather than evaluating performance outcomes, the study focuses on how guidance modalities influence sensation, cognition, and experience during indoor bouldering, a form of climbing performed on 3–5 meter walls without ropes or harnesses. By comparing auditory and visual feet guidance in a controlled indoor climbing setting, this work contributes to SportsHCI research by examining how multimodal interfaces can support experience-oriented movement practices among amateur athletes.

Accordingly, this study addresses the following research questions:

RQ1: How do multimodal interfaces for feet guidance affect proprioception and spatial awareness during bouldering?

RQ2: Can multimodal interfaces for feet guidance influence climbers' bouldering experience?

Related Work

HCI Research in Climbing

Previous HCI research has explored climbing as a testbed for interactive systems that combine physical movement with technological augmentation. Kosmalla et al. (2017) and Tiator et al. (2018) proposed mixed-reality climbing systems that integrate real climbing walls with virtual overlays, primarily to simulate high-risk scenarios and support psychological training. In the context of assistive technology, Bufalino (2022) and Ilich (2008) developed climbing guidance systems for visually impaired users, highlighting the potential of alternative modalities such as sound and vibration in environments where visual information is limited. More recent work has extended auditory guidance to broader user groups in bouldering contexts. Some studies have empirically demonstrated the feasibility of audio-based guidance for supporting climbing tasks or performance (Sun et al., 2025), while others have explored sonification as a design space for shaping climbers' aesthetic and experience (Yilmazer et al., 2025).

While these systems provide valuable contributions to performance enhancement, accessibility, and design exploration, empirical systems and evaluations have largely prioritized measurable efficiency, task completion, or physical outcomes. In contrast, the *Grand Challenges in SportsHCI* (Elvitigala et al., 2024) call for a shift away from performance-centric metrics toward designs that foreground embodied experience, playful exploration, and diverse personal goals. In particular, the authors emphasize the importance of supporting curiosity-driven, improvisational, and non-competitive forms of movement, especially for amateur and recreational participants.

A limited number of climbing-related studies have begun to address these experiential dimensions. Mencarini et al. (2016, 2018) showed that learning to climb involves emotional regulation, particularly in managing fear and building trust. Kiewa (2001) identified a sense of control as a central component of the climbing experience. Other studies have demonstrated that psychological training programs improved women climbers' performance when facing the fear of falling (Garrido-Palomino & España-Romero, 2023). However, cognitive dimensions, such as how climbers perceive their body position and spatial relations during movement, remain underexplored. Such topics are only approached indirectly through environmental or task design (Seifert et al., 2015) or through conceptual discussions and design guidelines (Yilmazer et al., 2025). This gap motivates the present study's focus on proprioception and spatial awareness as key experiential dimensions shaped by different guidance modalities.

Proprioception, Spatial Awareness, and Flow Experience in Sport

While prior HCI research in climbing has focused on system design, performance support, and accessibility, understanding how such systems affect climbers' embodied experience requires engaging with broader research on bodily perception and awareness in sport.

Proprioception

Proprioception, defined as the sense of limb position and movement, is fundamental to motor control and coordination in sport (Han et al., 2016). Sports science research has demonstrated that proprioceptive training can enhance balance, coordination, and movement efficiency (Yilmaz et al., 2024; Federici et al., 2020). In HCI, proprioception has increasingly been conceptualized as an interaction resource rather than a purely physiological capacity, showing how interfaces can leverage users' bodily awareness to provide guidance and feedback beyond visual displays (Lopes et al., 2015; Chen et al., 2014). However, comparatively little attention has been paid to how different guidance modalities reshape proprioceptive experience during sports activities. Prior research suggests that proprioception plays a critical role in shaping movement experience: excessive explicit attention to bodily movements may disrupt automatic motor processes, whereas more implicit or pre-reflective proprioceptive awareness supports fluent and adaptive action (Esmaeilinasab & Adnani, 2021). This distinction is particularly relevant in SportsHCI, where interactive systems may redirect athletes' attentional focus and thereby qualitatively shape different sport experiences.

Spatial Awareness

While proprioception concerns awareness of one's own bodily movement, spatial awareness extends this perception to the relationship between the body, external targets, and the rest of the environment. Prior research has shown that spatial awareness and visual-motor coordination are closely linked to skill performance in sport, yet spatial perception itself is not exclusively visual (Garg & John, 2025). Multisensory cues, including auditory feedback, can contribute to the perception of relative distance and orientation, particularly when visual information is constrained, e.g., Guarese et al. (2024) demonstrated that pitch-based sonification can effectively support spatial orientation in augmented reality contexts. From an embodied perspective, spatial awareness is actively constructed through movement and exploration rather than passively perceived, emerging from the dynamic interaction between the body and its environment (Tversky, 2009). Despite its key role in movement-based activities, how spatial awareness is shaped through different feedback modalities during real-world sport practice remains underexplored in Sports HCI research.

Flow Experience

Research on sport-oriented interaction design has increasingly argued that technologies should support athletes' authentic engagement with movement rather than distract from or dominate the activity itself. Prior work in Sports HCI emphasizes that effective designs should remain closely connected with ongoing bodily action, providing subtle, real-time, and non-intrusive feedback that allows athletes to stay attuned to their own movement and the surrounding environment (Mencarini et al., 2018; Nylande & Tholander, 2014).

These design principles resonate with the concept of flow, a state of deep engagement characterized by a merging of action and awareness, a sense of control, and intrinsic enjoyment, which has been extensively studied in both game design and sport psychology (Swann, 2016; Chen, 2007; Csikszentmihalyi and Rathunde, 1993). While HCI research in games has actively adopted flow as a design goal (Nijholt et al., 2008), Sports HCI research

has paid comparatively less attention to how interaction modalities shape athletes' subjective experience during real-world movement. Although prior work has explored the possibility of detecting flow states in sport contexts (Havlucu et al., 2018), relatively few studies have examined how interaction design itself may actively shape flow experience or support athletes in attaining such experiential states during practice. Notably, in this paper, flow is not a formally measured psychological state, but rather a conceptual lens for understanding qualities of engagement and embodied experience in climbing.

Methods

Formative Study

Before defining the focus of the main study, a formative study was conducted to understand which aspects of climbing experience climbers themselves consider most critical during bouldering. Semi-structured interviews were carried out with five climbers with varying levels of experience.

Across participants, difficulties related to body positioning and spatial judgment emerged as central challenges during climbing. Climbers frequently described uncertainty in judging distances, finding stable footholds, and positioning their bodies efficiently on the wall. While less experienced climbers tended to focus on bodily capability and confidence, more experienced climbers emphasized movement efficiency and precise body positioning.

When discussing potential assistance during climbing, participants expressed interest in support that could help them judge distance and find stable body positions while climbing. Importantly, climbers emphasized that such assistance should not overly prescribe actions or interrupt the climbing flow, but instead support exploration and allow climbers to remain in control of their movement decisions.

Findings from the formative study directly informed the design of the main experiment. The emphasis on body positioning and distance judgment motivated a focus on foot guidance, as foothold placement is often visually occluded and critical for stable movement. The repeated mention of bodily and spatial awareness guided the selection of proprioception and spatial awareness as key outcome measures. Finally, climbers' preference for supportive but non-directive assistance informed the choice of auditory guidance, which was designed to provide spatial cues without interfering with climbers' habitual movement patterns.

Guidance Modalities

This study compared two feet-guidance modalities: a traditional visual guidance condition and a spatial audio-based guidance condition. The selection of these two modalities was motivated by the aim to contrast established climbing instruction practices with a novel multimodal guidance approach, in order to examine how different modalities shape climbers' proprioception, spatial awareness, and overall climbing experience.

The visual guidance condition reflected common instructional practice in indoor climbing, where instructors use a laser pointer (or a wall-integrated LED system called Kilter Board, see Figure 1) to indicate the next target foothold. Climbers relied primarily on visual perception to locate the foothold and adjust their body position accordingly. The auditory guidance condition was developed based on prior work on sound-assisted climbing (Sun et al., 2025) and employed spatial audio cues to guide foot placement. In this condition, climbers were encouraged to use both auditory and visual information to interpret guidance and coordinate movement. The experiment focused specifically on foot guidance, as footholds often provide limited visual cues due to body occlusion and viewing angles, making modality-related differences more salient.



Figure 1: A Kilter Board system used in indoor climbing, with LED lights indicating the route and target holds to provide visual guidance.

(Picture source: <https://awesomewalls.co.uk/how-to-climb-a-kilter-board>)

Auditory guidance was generated based on the relative spatial relationship between the target foothold and the climber's foot that was closest to the target, as shown in Figure 2. Let the closest foot position be $F = (x_f, y_f)$ and the target foothold be $T = (x_t, y_t)$.

Three auditory parameters were used. First, the horizontal direction (X-axis) was encoded through stereo balance using the horizontal difference $\Delta x = x_t - x_f$. Δx was clamped to $[-2m, 2m]$, which was empirically determined during on-site testing to be both audible and distinguishable, and mapped to the stereo pan predefined to $[-1 \text{ (left channel)}, 1 \text{ (right channel)}]$.

The mapping was implemented as:

$$panStereo = Lerp(-1, 1, inverseLerp(-2, 2, \Delta x)).$$

When the target foothold was located to the right of the closest foot, the sound was weighted toward the right audio channel, gradually equalizing as the foot aligned horizontally with the target.

Second, horizontal distance (X-axis) was conveyed through tempo modulation based on horizontal difference $\Delta x_{abs} = |x_t - x_f|$. The distance was clamped to $[0.01\text{ m}, 15\text{ m}]$, which was empirically determined during on-site testing to be both audible and distinguishable, and mapped to the audio tempo predefined to $[0.1, 7]$ beeps per second.

The mapping was implemented as:

$$tempo = Lerp(0.1, 7, inverseLerp(15, 0.01, \Delta x_{abs})).$$

A faster tempo indicates closer proximity and a slower tempo indicates greater distance on the X-axis.

Third, vertical information (Y-axis) was conveyed only as a relative relationship (above vs. below), using a simplified pitch mapping based on vertical difference $\Delta y = y_t - y_f$. The pitch of the audio clip (ordinary pitch level $\sim 380\text{ Hz}$) was coded as a binary relation with Δy . If $\Delta y \geq 0$, the pitch was set to 760 Hz (2 times the ordinary level), indicating that the target foothold was above the foot; if $\Delta y < 0$, the pitch was set to 266 Hz (0.7 times the ordinary level), indicating that the target foothold was below the foot. These pitch levels were empirically determined during on-site testing to be both audible and distinguishable.

When the climber's foot position coincided with the target rock position in both axes, a confirmation sound effect was played.

Importantly, continuous change in vertical distance was not encoded given the physical and attentional demands of climbing, particularly for amateur climbers. This design choice was informed by prior research showing that fine-grained pitch-to-height mappings can impose increased cognitive load and require users to memorize and interpret subtle pitch variations (Guarese et al., 2024).



Figure 2: Illustration of the auditory feet-guidance mechanism. The blue sphere symbolizes the position of the target foothold (x_t, y_t) , and the red sphere symbolizes the position of the closest foot (x_f, y_f) . The auditory parameters change based on the relative position Δx and Δy .

Apparatus

The experiment employed the same apparatus as the previous study (Sun et al., 2025). A Microsoft Xbox One Kinect sensor was used to track climbers' full-body movements in real time and was integrated with the Unity engine via the Kinect MS-SDK. The Unity engine processed the tracking data, managed the spatial configuration and sequence of target footholds, and dynamically controlled the auditory feedback during climbing. Participants

wore wireless stereo headphones to receive the audio guidance (see Figure 3).

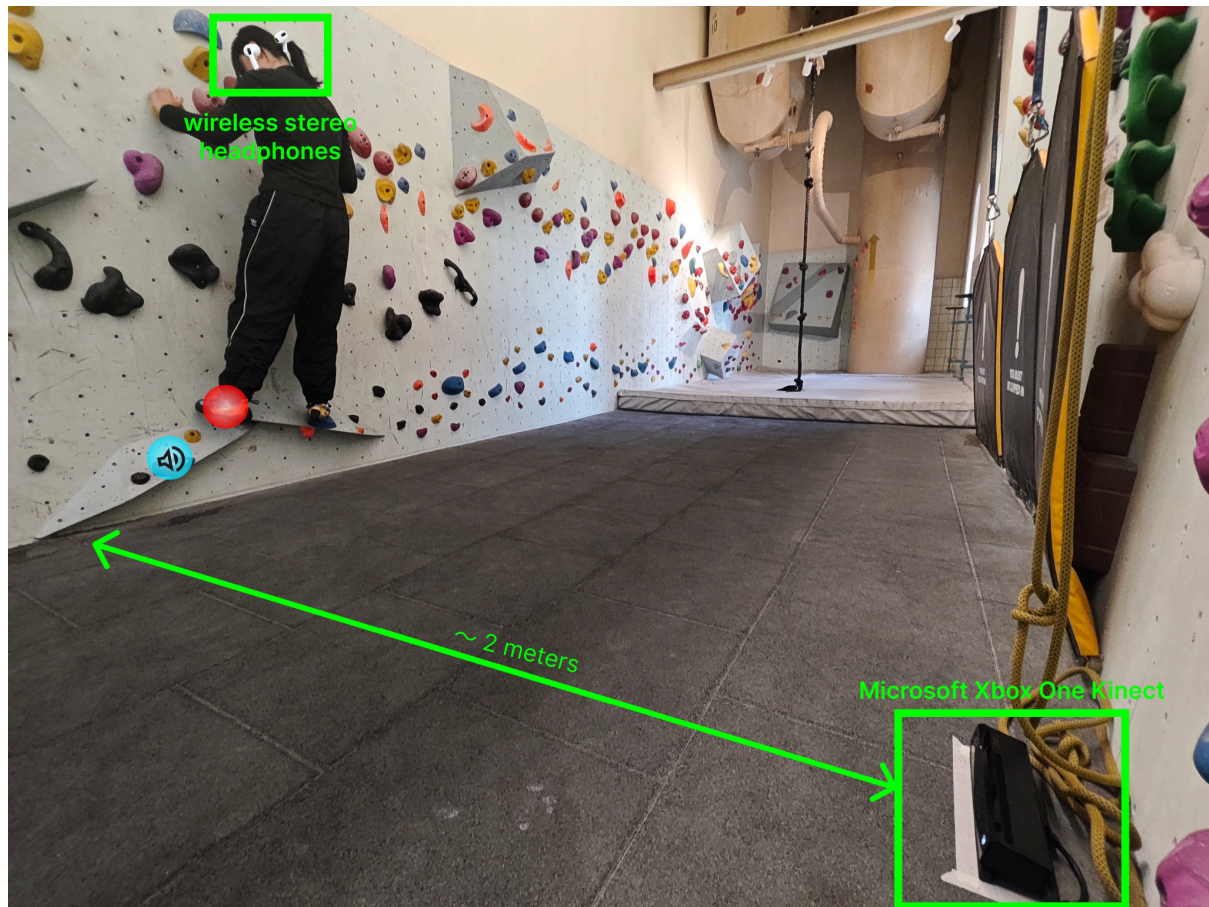


Figure 3: Experimental setup for the auditory feet-guidance condition. A climber wears wireless stereo headphones while performing a bouldering route, with full-body motion tracked by a Microsoft Xbox One Kinect positioned approximately 2 meters from the wall.

Experimental Setup

The experiment was conducted on a horizontal bouldering wall in an indoor climbing gym, consistent with the previous study (Sun et al., 2025). Two climbing routes were pre-defined and mapped in the Unity 3D environment. Each route consisted of nine target footholds and was designed to be equivalent in overall physical displacement and difficulty. Participants were instructed to touch the target footholds sequentially with their feet, moving from the start position to the end position (see Figure 4). Both routes were beginner-friendly and prioritized participant safety. All trials were conducted during periods when the climbing area was clear and quiet to minimize external distractions.

The study employed a within-subjects design, in which all participants completed both guidance modalities. Participants were recruited both offline at the climbing gym and online through climbers' group chats. To control for order effects and route-specific influences, both the presentation order of guidance modalities and the pairing between routes and modalities were fully counterbalanced across participants. Specifically, half of the participants completed Route A with visual guidance and Route B with auditory guidance, while the other half completed Route A with auditory guidance and Route B with visual guidance.



Figure 4: 3D scan of Route A (left) and Route B (right). Star icons symbolize the target footholds climbers were guided to touch with their feet, the start position and the end position marked. Arrows demonstrate the prescribed order of the targets along each route.

Experimental Procedure

Prior to the experiment, participants were asked to complete a consent form. Then they filled in a demographic questionnaire on registration. The questionnaire collected information on age, gender, climbing experience and frequency, primary climbing discipline, self-reported climbing level, as well as any hearing or vision-related impairments. Participants were also asked about prior experience with auditory instructions or musical training.

On-site, participants completed two climbing rounds following the predefined route, with modality pairing and order assigned to them. After completing each climbing round, participants immediately filled out a post-task questionnaire consisting of the System Usability Scale (SUS; Brooke, 1996), the NASA Task Load Index (NASA-TLX; Hart & Staveland, 1988), and a set of original 5-point Likert-scale items assessing proprioception, spatial awareness, and overall climbing experience as supplementary to qualitative findings (see Table 1).

Metric	Items
Proprioception	(1) "I felt aware of where my foot was at all times." (2) "I felt aware of how my foot was moving throughout the climb."
Spatial awareness	(1) "I felt aware of where I was in relation to my surroundings." (2) "I felt aware of where the target foothold was in relation to my body."
Climbing experience	(1) "I really enjoyed the climbing process." (2) "I felt motivated during the climb." (3) "I felt willing to explore more during the climb." (4) "I felt confident about future climbing challenges."

Table 1: Original designed Likert scale items answered by participants to assess their Proprioception, Spatial Awareness, and Climbing Experience after each trial.

On completion of both climbing rounds, a short semi-structured interview was conducted with each participant. The interview focused on participants' comparative experiences between the two guidance modalities. Participants were also invited to provide general

feedback and reflections on both guidance systems. All interviews were audio-recorded and later transcribed for analysis.

Results

A total of twelve participants took part in the study. The sample consisted predominantly of female climbers (83.3%), with two participants identifying as male (16.7%). Participants ranged in age from 23 to 31 years ($M = 27.0$).

Regarding climbing experience, five participants (41.7%) reported at least one year of climbing experience, four participants (33.3%) reported between six and twelve months of experience, and three participants (25.0%) reported less than six months of experience. Weekly climbing frequency was relatively high, with six participants (50.0%) climbing once per week and five participants (41.7%) climbing two or more times per week.

In terms of climbing discipline, bouldering was the most commonly reported primary climbing type (58.3%), followed by top-rope climbing (41.7%).

With respect to sensory-related background, one participant reported mild hearing difficulties that did not affect daily activities. No participants reported color blindness or other vision-related impairments. Participants' prior experience with musical training or auditory cue-based activities varied, with most participants reporting either no prior experience (41.7%) or a neutral level of familiarity (33.3%).

A detailed overview of participant demographics and climbing background is provided in Table 2.

partici pant ID	How old are you?	What is your gender?	How long have you been climbing?	How often do you climb in a week?	Which type of climbing are you most experienced with?
1	28	Male	Between 6-12 months	2 times or more	Top-rope climbing
2	25	Female	Less than 6 months	2 times or more	Bouldering
3	23	Female	Less than 6 months	1 time	Top-rope climbing
4	31	Female	1 year or longer	1 time	Bouldering
5	28	Male	Between 6-12 months	2 times or more	Bouldering
6	30	Female	Between 6-12 months	1 time	Top-rope climbing
7	28	Female	1 year or longer	2 times or more	Bouldering
8	25	Female	Less than 6 months	less than 1 time	Bouldering
9	24	Female	1 year or longer	1 time	Top-rope climbing
10	25	Female	1 year or longer	1 time	Top-rope climbing
11	27	Female	1 year or longer	1 time	Bouldering
12	29	Female	Between 6-12 months	2 times or more	Bouldering

Table 2: Demographic characteristics and climbing backgrounds of the participants.

System usability and task workload

To provide an overview of participants' general experience with the two guidance modalities, overall system usability and perceived workload were first assessed using the System Usability Scale (SUS) and the NASA Task Load Index (TLX).

SUS responses were scored following the standard procedure described by Brooke (1996), resulting in a composite usability score ranging from 0 to 100. For the NASA-TLX, ratings on the six subdimensions—mental demand, physical demand, temporal demand, perceived performance, effort, and frustration—were analyzed using the raw TLX approach, in which subscale ratings are treated independently rather than combined through weighted aggregation (Hart & Staveland, 1988).

The visual interface received relatively high usability ratings ($M = 77.08$, $SD = 8.31$). In contrast, the auditory interface was rated notably lower in usability ($M = 57.08$, $SD = 18.12$). The greater variability in SUS scores for the auditory condition suggests considerable differences among participants in how they understood and adapted to the sound-based guidance.

Examination of the TLX subscales revealed clear modality-related differences in workload profiles (Table 3). The largest differences between conditions were observed in mental demand, physical demand, and frustration, all of which were higher under auditory guidance. Among these dimensions, frustration also showed substantially greater variability in the auditory condition than in the visual condition, indicating larger individual differences in affective response to sound-based guidance.

In contrast, temporal demand differed only slightly between modalities, suggesting that auditory guidance did not substantially increase perceived time pressure. Effort was also higher in the auditory condition, although the difference was smaller than that observed for mental and physical demand.

subdimensions	visual - mean	visual - sd	sound - mean	sound - sd
mental demand	2.08	1.08	3.67	1.78
physical demand	2.42	1.00	3.92	1.24
temporal demand	2.83	1.11	3.17	1.34
performance	6.17	0.94	5.50	0.80
effort	3.17	1.27	4.00	1.41
frustration	1.83	0.83	3.17	1.64
Average TLX score*	2.4	0.62	3.4	0.78

Table 3: Mean and standard deviation of NASA-TLX subdimensions for visual and auditory guidance conditions. (*Performance is positively framed, therefore was inverted before computing the average TLX score)

Lower usability and higher workload in the auditory condition do not necessarily indicate poorer experience overall. Rather, these results suggest that auditory guidance engages

climbers in a qualitatively different manner, placing distinct demands on the climbing process, as reported in the following sections.

Proprioception

Participants (P3, P5, P9, P10, P12) frequently reported an enhanced sense of proprioception under audio-guided conditions. This perceived enhancement appeared to be closely related to the way audio feedback was directly coupled with foot movement, which required participants to actively map changes in sound to their bodily position. As a result, more attentional resources seem to have been allocated to the feet and lower-body movements, as participant P10 described:

“Because you do not know which foothold the sound is referring to, you end up lifting your leg and trying to reach in different ways. Sometimes you are not sure whether you should go higher, so you lift your leg more. Through this process, you can clearly feel the stretching and movements of your body.” (P10)

However, several participants also reported (P3, P8, P11) that under audio-guided conditions, their body positioning felt less efficient and therefore more physically tiring. This was mainly because they needed to first follow and interpret the audio instruction, then settle their feet through adjustment, and only afterwards find a stable and comfortable posture. In contrast, under visual guidance, they were able to plan their posture and movements more quickly and efficiently:

“With visual guidance, you can adjust yourself into a more efficient posture. With audio guidance, you reach out, stretch your arms, look farther away, and then try to find a more comfortable position.” (P3)

Under visual guidance, participants (P6, P10, P12) often described their movements as being driven primarily by external cues rather than internal bodily sensations. Instead of actively sensing their own body position, they tended to rely on visual information to execute movements directly and efficiently:

“With the laser pointer, you do not really feel your feet. You first look for the point with your eyes, and then you simply move your body there.” (P12)

These findings suggest that audio guidance enhances proprioceptive awareness by interrupting habitual, visually driven body movement, while visual guidance makes bodily sensations more implicit and less conscious. Questionnaire results on proprioception showed comparable levels of reported proprioceptive awareness across modalities, with slightly higher mean scores in the visual condition ($M = 3.92$, $SD = 0.85$) than in the auditory condition ($M = 3.71$, $SD = 0.62$), although these differences did not prove to be statistically different. This mismatch will be further discussed in the Discussion section.

Spatial Awareness

Participants (P1, P11) reported that spatial awareness during climbing relied on the coordination of multiple sensory modalities, particularly a combination of visual and auditory

information. Even under audio-guided conditions, participants continued to use visual cues to support their spatial understanding. Audio guidance was described as helpful for indicating relative spatial relationships rather than absolute positions, as participant P7 explained:

“Audio feedback gives you a sense of relative position.” (P7)

Accordingly, spatial awareness ratings were highly comparable between the two conditions, with mean scores of 3.71 (SD = 0.58) for the visual modality and 3.67 (SD = 0.44) for the auditory modality, with no statistically significant difference being found between them. This similarity suggests that both guidance modes supported basic spatial orientation to a similar degree.

Interestingly, several participants (P4, P5, P7) reported that the routes felt more dynamic and spatially exaggerated under audio guidance, particularly along the vertical dimension. Although the two routes were designed to be equivalent in physical displacement and difficulty and were counterbalanced across conditions, participants perceived the audio-guided route as involving greater vertical variation, whereas the visually guided route was experienced as emphasizing more horizontal movement:

“The first (audio guidance) route felt more varied, with much larger vertical changes. The second one (visual guidance) felt more horizontal.” (P7)

At the same time, participants also reported (P1, P7, P9) limitations in their ability to fully perceive spatial information through audio guidance alone. In particular, some participants found it difficult to distinguish vertical differences based on sound:

“I could not really hear any difference between going up and going down.” (P1)

The perceived differences between audio and visual modalities were not reflected in the quantitative spatial awareness results, indicating no clear modality effect in self-reported spatial awareness. These findings suggest that while audio guidance can enrich spatial awareness by emphasizing relative distance and amplifying certain spatial dimensions, it does not fully replace visual perception. Instead, spatial awareness during climbing emerges from the integration of multiple sensory inputs, with audio cues playing a complementary but constrained role.

Climbing Experience

Participants generally reported a positive climbing experience under both audio and visual conditions. However, the audio condition was more frequently described as interesting and enjoyable, and was often compared to playing a game. In particular, participants (P2, P4, P5, P6, P8, P11) appreciated the gamified nature of the auditory feedback, especially when successfully reaching a target foothold and receiving immediate sound confirmation:

“It did not feel like I was climbing. It felt more like playing a small game that gives you immediate feedback, which was very enjoyable.” (P8)

At the same time, several participants (P3, P4, P5) also described the audio condition as more challenging. This challenge was partly attributed to the need for repeated attempts and exploratory movements in order to interpret the sound cues, which could lead to increased physical effort and fatigue. Such feedback could be proved by higher TLX physical demand subscores under the audio condition.

Several participants (P6, P9, P12) emphasized that the audio mode afforded a stronger sense of autonomy than the visual mode, as it allowed climbers to regulate their own climbing pace rather than waiting passively for an external instruction. As one participant explained:

“It feels like it gives me more autonomy. I can look and decide by myself, instead of just waiting there.” (P12)

In contrast, the visual mode was sometimes described (P2, P4, P7, P10) as more controlling and directive, resembling a task assigned by an external authority:

“It feels like a teacher giving you a task and telling you exactly what the next step is.” (P4)

This difference in perceived autonomy was closely related to a broader experiential distinction between the two modes. Participants often characterized the audio condition as an exploratory and discovery-oriented process, in which the guidance provided orientation rather than a fixed goal. Under audio guidance, climbers described being less explicitly goal-driven and more engaged in exploration:

“With sound, it only gives you guidance at the same time. I am not very goal-oriented about my next move, and I spend more time exploring.” (P10)

Importantly, this qualitative pattern was also reflected in the descriptive quantitative results. Participants reported higher exploration willingness under auditory guidance than under visual guidance, with a mean score of 4.17 (SD = 0.83) in the auditory condition compared to a mean score of 3.75 (SD = 0.97) in the visual condition, although this difference was not found to be statistically significant, likely due to the low sample size. This descriptive difference aligns with participants’ interview accounts describing greater exploratory engagement when guided by sound.

The findings suggest that auditory cues may encourage climbers to engage more actively with the climbing surface and to experiment with their own bodily strategies. Unlike laser-based visual guidance, which provides a precise and externally defined target point, auditory feedback was less deterministic and required climbers to interpret spatial cues through continuous bodily adjustment and decision-making. While this tendency is evident in both participants’ narratives and the descriptive statistics, it reflects a shift in experiential orientation rather than a definitive performance advantage.

Learning Load & Mode Familiarity

Almost all participants reported that the audio mode required more time and practice to become familiar with, compared to the visual mode. Several participants (P1, P4, P5, P7,

P8, P12) indicated that additional exposure or repeated trials would be necessary in order to understand and adapt to the different sound signals used for guidance:

“Maybe I need another round of the experiment to get myself familiar with the different sound signals.” (P4)

Participants (P5, P6, P11) also emphasized that interpreting the mapping between sound cues and movement directions involved greater learning effort or cognitive processing than visual guidance. Unlike visual cues, which could be acted upon immediately, auditory cues required participants to pause, mentally process the information, and then test their movements incrementally:

“I need to wait and let my brain process the information first. After I finish that judgment, I know whether to move left or right, then I try, adjust again, and try once more.” (P5)

In contrast, the visual mode was consistently described as more intuitive (P1, P4, P7). Participants reported that they were accustomed to relying on visual information during climbing.

This pattern was also reflected in subjective workload ratings. This pattern was also reflected in the mental demand subscale of the NASA Task Load Index. Participants reported higher perceived mental demand under the auditory condition ($M = 3.67$, $SD = 1.78$) than under the visual condition ($M = 2.08$, $SD = 1.08$), aligning with interview accounts describing increased cognitive effort when interpreting sound-based guidance.

These findings suggest that while the audio mode introduced a higher learning load and required deliberate practice to master, the visual mode aligned more closely with climbers' existing habits and experience, resulting in a more immediately accessible form of guidance.

Discussion

Addressing the Research Questions

This experiment investigated how visual and auditory feet-guidance modalities influence proprioception, spatial awareness, and subjective climbing experience in amateur climbers. The findings suggest that rather than improving proprioception or spatial awareness in a measurable sense, auditory guidance reshaped how climbers attended to and constructed bodily and spatial information during climbing.

Audio guidance tended to disrupt habitual, visually driven climbing routines, prompting greater attentional focus on foot movements and exploratory decision-making. This was associated with stronger experiential awareness and sense of autonomy, but also with increased learning demands and physical effort. In contrast, visual laser-based guidance aligned closely with climbers' existing perceptual habits, enabling more efficient posture planning and lower cognitive workload, while rendering bodily sensations more implicit.

The mismatch between the quantitative and qualitative results is interesting. One possible explanation is that the original questionnaire, which was intended to supplement interview

findings, was not a fully validated instrument for measuring proprioception. Beyond measurement limitations, this discrepancy may also reflect a distinction between *explicit* and *implicit* proprioceptive awareness, as discussed by Esmailinasab & Adnani (2021). Here, 'explicit' refers to the conscious cognitive effort required to interpret auditory cues, whereas 'implicit' refers to the pre-reflective bodily engagement that emerged through continuous exploratory movement. Auditory guidance may have simultaneously fostered a more implicit, embodied form of proprioceptive engagement through exploratory movement, whereas visual guidance supported more readily reportable, explicit bodily awareness.

Participants' reports of exaggerated vertical movement under auditory guidance further indicate modality-specific differences in spatial experience. Audio cues emphasized relative spatial relationships and bodily exploration, whereas visual guidance supported direct visual estimation. Design constraints in the current audio mapping may also have limited the precision with which vertical distance could be conveyed. These experiential differences are examined in detail in the following discussion sections.

Exploration, Autonomy, and the Cost of Engagement

The findings indicate that auditory feedback guidance fostered a more exploratory and self-directed mode of climbing. Rather than prescribing a precise target, auditory cues provided continuous orientation that required climbers to actively interpret spatial information through bodily movement. This led participants to report a stronger sense of autonomy and engagement. Several climbers described becoming absorbed in responding to the sound feedback, focusing on movement exploration rather than on completing the route as efficiently as possible. Such experiences resemble key characteristics of a flow-like state, including sustained attention, intrinsic enjoyment, and a sense of control (Chen, 2007). This aligns with prior SportsHCI work suggesting that subtle and non-directive feedback can support authentic engagement with movement (Mencarini et al., 2018; Nylande & Tholander, 2014), and demonstrates how multimodal interfaces can be leveraged to enrich subjective sport experience beyond performance-oriented outcomes.

However, this increased exploration and autonomy came with a clear cost of engagement. Auditory guidance imposed higher cognitive and physical demands, as reflected in elevated mental and physical workload scores and participants' accounts of increased effort and fatigue. Interpreting sound cues required deliberate attention and learning, particularly for climbers accustomed to visually driven climbing strategies. This trade-off highlights an important design tension: while auditory guidance can enrich the climbing experience by fostering exploration and self-directed action, it may also challenge efficiency and ease of use, especially in early encounters. Rather than representing a drawback, this cost can be understood as an inherent characteristic of high-engagement designs that prioritize experiential depth over immediate performance gains, a distinction that is central to recent discussions in SportsHCI (Elvitigala et al., 2024).

System Design and Usability Drawbacks

Several limitations in the system design and experimental setup should be considered when interpreting the findings. To ensure comparability between conditions, the experiment

simplified some aspects of real climbing practice, such as ignoring foothold color as a visual cue. Participants frequently noted that the predefined routes and guidance-based task structure felt different from natural climbing, which reduced the sense of authentic sport experience and made the task feel more game-like. This simplification may have influenced how climbers perceived effort and movement intention, especially under the auditory condition.

Technical limitations of the sensing setup also affected the usability of the auditory guidance. The Kinect-based tracking system occasionally showed a limited field of view, latency, or inaccurate foot detection, which interrupted the continuity of sound feedback. These issues likely contributed to the higher workload, frustration, and greater variation in usability ratings reported for the auditory condition. Future work could improve this by using more accurate body-tracking models or mobile sensing devices.

Finally, due to time and space constraints, participants were given only three practice points before the main trials. Several participants reported that this was not enough to fully understand and get used to the sound mappings. As a result, some of the differences between modalities may reflect differences in familiarity rather than inherent experiential qualities. Future studies could include longer practice periods or repeated use to better separate learning effects from modality-related experience differences.

Conclusion

This study investigated how different feet-guidance modalities shape amateur climbers' proprioception, spatial awareness, and subjective climbing experience. By comparing a traditional visual laser-based guidance system with a spatial audio-based guidance approach in an indoor bouldering setting, the study examined how multimodal feedback influences climbers' bodily perception, spatial understanding, and engagement during movement. Through a mixed-methods approach combining questionnaires and post-task interviews, the findings highlight how auditory guidance alters attentional focus and experiential orientation in climbing, offering insights into the role of multimodal interfaces in shaping embodied sport experiences. Future work could examine longer-term use of auditory guidance, explore more naturalistic climbing settings, and investigate how multimodal feedback can be tailored to different skill levels and climbing styles.

Appendices

Interview Outline in Formative Study:

1. Do you pursue a better performance in climbing, or do you climb just for fun?
2. What are the most important factors that usually affect your climbing performance?
3. In which phases of climbing (before, during, or after) do you feel the need for assistance, and what kind of assistance do you expect in each phase?
4. If there were a technology that could support you in climbing, would you expect it to help with the following aspects? How do you expect it and why?
5. Considering everything we talked about, did anything new come to your mind?

Interview Outline in Formal Study

1. How did you feel about the guidance system and the climbing experience in each round? Can you explain why?
2. How clear did you find the guidance in each round? What kinds of information did you notice while climbing (e.g., direction, distance)? How did you use this information to make movement decisions during the climb?
3. Did you notice any differences between the two round? Can you describe these differences?
4. Is there anything else you would like to share about your experience with the two guidance systems or the climbing task overall?

Questionnaire Before Climbing

1. How old are you?
2. What is your gender?
 - a. Female
 - b. Male
 - c. Non-binary / Gender diverse
 - d. Prefer not to say
3. How long have you been climbing? You can describe by yourself. Please specify total experience in years/months.
4. How often do you climb? You can describe by yourself. Please specify how many times a week/month.
5. Which type of climbing are you most experienced with?
 - a. Top-rope climbing
 - b. Lead climbing
 - c. Bouldering
6. Do you have any hearing difficulties?
 - a. No.
 - b. Yes, but it does not affect my daily activities.
 - c. Yes, and it affects my daily activities.
 - d. Prefer not to say
7. Do you have any color blindness or vision difficulties?
 - a. No.
 - b. Yes, but it does not affect my daily activities.
 - c. Yes, and it affects my daily activities.
 - d. Prefer not to say.
8. I have musical training or experience following auditory cues (e.g., gaming, sound-based navigation, or radar/sound direction). Please rate how well this statement describes you.
 - a. Strongly disagree
 - b. Disagree
 - c. Neutral
 - d. Agree
 - e. Strongly agree

Questionnaire After Climbing

SUS: To what extent do you agree with the description regarding the guided climbing system in this round? Please select the most suitable option.

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

NASA-TLX: To what extent do you agree with the description regarding the workload of the climbing task in this round? Please rate your feelings.

1. How mentally demanding was the task?
2. How physically demanding was the task?
3. How hurried or rushed was the pace of the task?
4. How successful were you in accomplishing what you were asked to do?
5. How hard did you have to work to accomplish your level of performance?
6. How insecure, discouraged, irritated, stressed, and annoyed were you?

Ordinary Items: To what extent do you agree with the description regarding the overall climbing experience in this round? Please select the most suitable option.

1. I felt myself aware of where I am in relation to my surroundings.
2. I felt myself aware of where the target rock is in relation to my body.
3. I felt myself aware of where my foot is all the time.
4. I felt myself aware of how my foot is moving all the time.
5. I really enjoyed the climbing process.
6. I felt motivated in the climbing process.
7. I feel more willing to explore more in the climbing process.
8. I feel confident about future climbing challenges.
9. How did the second climbing session feel compared to the first one?
10. Do you have other feelings about the system and the climb, or any other words to say? Feel free to tell us!

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